MILITARY SPECIFICATION

CAP, LUBRICATION FITTING, PROTECTIVE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers a reusable, nonmetallic type of cap for use in protecting lubrication fittings during sand blasting and painting operations.

2. APPLICABLE DOCUMENTS

2.1 Applicable documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification.

SPECIFICATIONS

FEDERAL

LLL-B-636 — Boxes, Fiber, Solid, Domestic.

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood Cleated.

PPP-B-601 — Boxes, Wood, Cleated Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-676 — Boxes, Set Up, Paperboard.

PPP-T-76 — Tape, Pressure-Sensitive, Paper, Adhesive, Water Resistant.

MILITARY

MIL-P-116 — Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MS24529 — Caps, Lubrication Fitting, Protective.

MS15000 — Fittings, Lubrication (Hydraulic) Acceptable Tip Designs.

(Copies of specifications or standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4730

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3. REQUIREMENTS

3.1 Design. Caps shall be designed in conformance with MS24529 and shall have sufficient elasticity to permit installation and removal without the use of tools.

3.2 Material. The caps shall be made of an oil and abrasion resistant elastomeric composition and shall have sufficient resiliency to resist erosion when used to protect fittings during sandblasting operations. Elasticity of the material shall be sufficient to prevent dislodging of the caps during sandblasting operations.

3.3 Oil resistance. The caps shall have sufficient oil resistance to pass the test specified in paragraph 4.3.1 without deforming, swelling, loss of elasticity or any undesirable change in the physical characteristics of the material.

3.4 Abrasion resistance. The caps shall be abrasion resistant. When tested as specified in paragraph 4.3.2 the loss in weight shall not exceed 5 percent of the original weight of the cap.

3.5 Color. The color shall be optional.

3.6 Workmanship. Workmanship shall be of high quality in order to produce uniform caps, free of bloom, flash, oil, or other harmful foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. A lot shall consist of all caps for one contract or order, produced from the same compound, manufactured under essentially the same conditions, and offered for inspection at one time but shall not exceed 100,000 caps.

4.2 Sampling for testing. Samples for testing (pars. 4.3.1 and 4.3.2) shall be taken by a government inspector in accordance with inspection level L2 of the appendix to MIL-STD-105 or as designated by the contracting officer.

4.2.1 *Sampling for inspection.* Samples for inspection of minor defects shall be taken by a government inspector in accordance with inspection level L8 of the appendix to MIL-STD-105 or as designed by the contracting officer.

4.3 Testing. Samples as selected in paragraph 4.2 shall be tested by Government inspectors or under government supervision at a laboratory designated by the contracting officer. The acceptable quality level shall be 2.5 percent.

4.3.1 *Oil resistance.* The test caps shall be immersed in SAE 10 lubricating oil at 140 ± 10° F. for 1 hour. The caps shall then be removed and inspected for deformation, swelling, loss of elasticity (4.4.1) or any other change affecting serviceability.

4.3.2 *Abrasion resistance.* The test cap shall be weighed to within 0.01 of a gram and mounted on a lubrication fitting. The abrasive material, such as, sharp sand, slag, silicone carbide or aluminum oxide shall be of the following sieve size.

	Specified limits
Retained on #8 mesh screen.....	0–0 percent
Retained on #12 mesh screen.....	0–5 percent
Retained on #16 mesh screen.....	55–100 percent
Retained on #40 mesh screen.....	0–35 percent
Passing on #40 mesh screen.....	0–2 percent

This abrasive shall be directed at the cap for 15 ± ¼ minutes on a 45-degree angle from the verticle with the orifice of the blast nozzle 3 ± ¼ inches from the nearest surface of the cap. The temperature of the abrasive shall be 25 ± 2° Centigrade. The air pressure shall be 85 ± 5 psi. The cap shall then be removed and weighed to within 0.01 of a gram and the percent loss in weight determined.

4.4 Inspection. Samples as selected in para-

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graph 4.2.1 shall be inspected at place of manufacture unless otherwise designated by the contracting officer.

4.4.1 *Elasticity.* Caps shall be inspected for conformance to paragraph 3.1 by installing on lubrication fittings conforming to MS 15000. The caps shall fit securely and shall be removable without the use of tools or undue effort.

4.4.2 *Workmanship.* Caps shall be visually inspected for conformance to paragraph 3.6.

4.4.3 *Dimensions.* The caps shall be inspected for dimensional conformance to MS 24529 —Caps, Lubrication Fitting, Protective.

4.5 **Packing and packaging.** Packing and packaging shall be inspected for conformance to section 5 and the level of protection specified in the contract or order.

4.6 **Classification of defects.** Inspection shall be conducted as follows:

Categories and defects

Major	*A.Q.L. 2.5 percent*	*Inspection method*
1. Inside diameter		Drill rod.
2. Elasticity (4.4.1)		Manual.
Minor	*A.Q.L. 6.5 percent*	*Inspection method*
51. Workmanship		Visual.
52. Dimensions other than inside diameter		Machinists scale.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* 100 ea caps shall be placed in paperboard boxes conforming to Specifications PPP-B-566 or PPP-B-676. The method of preservation shall be Method III of Specification MIL-P-116.

5.1.1.2 *Intermediate packaging.* Caps, packaged as described in 5.1.1.1 shall be placed in a close-fitting fiberboard intermediate container conforming to Specification PPP-B-636. Closure shall be in accordance with box specification. Tape for closure shall conform to Specification PPP-T-76, 2-inch width minimum. Gross weight of package shall not exceed 20 pounds.

5.1.2 *Level B.* See section 6.

5.1.2.1 *Unit packaging.* Shall be as described in paragraph 5.1.1.1.

5.1.2.2 *Intermediate packaging.* Caps, unit packaged as described in 5.1.1.1 shall be placed into close-fitting fiberboard intermediate containers conforming to Specification PPP-B-636. Closure shall be in accordance with box specification. Gross weight of package shall not exceed 20 pounds.

5.1.3 *Level C.* Unit and intermediate packaging shall be in accordance with the manufacturers best commercial practice.

5.2 Packing.

5.2.1 *Level A.* Intermediate packages as described in 5.1.1.2 shall be packed into snug fitting exterior boxes conforming to Specifications PPP-B-621, Class 2, PPP-B-601, overseas type, PPP-B-585, Class 3 or PPP-B-591, overseas type. The box shall be strapped in accordance with applicable box specification. Gross weight of box shall not exceed 200 lbs.

5.2.2 *Level B.* Intermediate packages as described in 5.1.2.2 shall be packed into snug-fitting boxes conforming to Specifications PPP-B-61, Class 1, PPP-B-601 domestic type PPP-B-591, domestic type or PPP-B-585, Class 1. Gross weight of box shall not exceed 200 lbs.

5.2.3 *Level C.* Caps, packaged as described in 5.1.3 shall be packed to afford protection against damage during direct shipment from

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the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 **Marking.** Unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL–STD–129. All Level A exterior container markings shall be protected with a coating of transparent waterproof material in accordance with MIL–STD–129.

5.3.1 *Special markings.* Exterior containers inclosing level B or level C packages shall be marked LTD–MIL–PKG or MIN–MIL–PKG as applicable.

6. NOTES

6.1 **Intended use.** Caps purchased under this specification are intended for use as a protective device for lubrication fittings during sandblasting and painting operations.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Where inspection will be conducted (par. 4.3).

(c) Level of protection required.

(d) Inspection level if other than specified (pars. 4.2, 4.2.1).

1. Level B preservation and packaging is intended to provide economical but limited protection, and should be specified only when it is determined to be appropriate. Effective period of protection 1 year from date of initial packaging.

6.3 **Storage surveillance.** Item preserved and packaged in accordance with level B requirements must be inspected to determine condition when not used within the time period indicated. Items not used within the time period specified must either be preserved or repackaged in accordance with level B requirements in this specification or with level A requirements if storage beyond an additional year is anticipated.

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Other interests:
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